Test Plan Document

AI Proposal Generator

Prepared By:

M.G.S. Sewmini Jayasinghe (26063) - Quality Assurance

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1. Introduction

1.1Purpose

This Test Plan document describes the testing strategy, scope, approach, resources, schedule, and deliverables for the **AI Proposal Generator System**. It ensures that the system meets all functional and non-functional requirements and delivers high-quality AI-generated proposals.

1.2 Project Overview

The AI Proposal Generator is a web-based application that automatically generates structured business proposals using Artificial Intelligence based on user-provided inputs such as project details, requirements, and constraints.

The system aims to:

- Reduce manual effort in proposal writing
- Improve response time to client requests
- Maintain consistency and professionalism in proposals

2. Objectives

The objectives of testing are:

- Validate system functionality against requirements
- Ensure accurate and structured AI-generated proposals
- Verify proper integration of frontend, backend, and AI services
- Identify and report defects
- Ensure system stability, performance, and usability

3. Test Scenarios

🔷 **Module M2: Input Form UI**

TS-01: Verify all input fields are displayed (Client Name, Company Name, Project Title, Description, Features, Budget, Timeline, Platform Type)
TS-02: Verify user can enter data into all fields
TS-03: Verify UI layout and alignment of form fields
TS-04: Verify responsive design (desktop & mobile view)
TS-05: Verify dropdown selection for platform type

TS-06: Verify placeholder text and labels are correct
TS-07: Verify form accepts multi-line input for project description
TS-08: Verify maximum input length handling

🔷 Module M3: Form Validation

TS-09: Verify required fields cannot be submitted empty
TS-10: Verify error messages for missing required fields
TS-11: Verify optional fields (Budget, Timeline) can be skipped
TS-12: Verify input sanitization (no script injection)
TS-13: Verify special characters handling
TS-14: Verify validation messages disappear after correction
TS-15: Verify form submission with valid inputs only

🔷 Module M4: Backend API

TS-16: Verify API receives correct payload from frontend
TS-17: Verify API handles missing fields properly
TS-18: Verify API returns correct success response (200)
TS-19: Verify API returns error response for invalid request (400)
TS-20: Verify API handles server errors (500)
TS-21: Verify API timeout handling
TS-22: Verify API key authentication (if applicable)

🔷 Module M5: AI Integration

TS-23: Verify AI generates proposal with valid input
TS-24: Verify AI response includes all required sections
TS-25: Verify AI response relevance to user input
TS-26: Verify AI handles vague input (e.g., “make proposal”)
TS-27: Verify AI handles large input data
TS-28: Verify system handles empty AI response
TS-29: Verify system handles malformed AI response
TS-30: Verify consistency of output for same input
TS-31: Verify prompt processing and response parsing

🔷 Module M6: Proposal Output View

TS-32: Verify proposal is displayed after generation
TS-33: Verify all sections are rendered correctly:

Introduction

Requirement Understanding

Proposed Solution

Features

Timeline

Budget

Conclusion

TS-34: Verify formatting and readability of proposal
TS-35: Verify long proposals are displayed properly (scrolling)
TS-36: Verify no UI break for large content
TS-37: Verify proper section separation and headings

🔷 Module M7: Editing & Copy

Editing

TS-38: Verify user can edit proposal sections
TS-39: Verify changes are saved after editing
TS-40: Verify no formatting issues after editing
TS-41: Verify editing does not break structure

Copy

TS-42: Verify copy-to-clipboard functionality
TS-43: Verify copied content matches displayed content
TS-44: Verify full proposal is copied (no truncation)

🔷 Module M8: PDF Export

TS-45: Verify PDF download is triggered successfully
TS-46: Verify PDF contains full proposal content
TS-47: Verify formatting in PDF matches UI
TS-48: Verify no missing sections in PDF
TS-49: Verify PDF download for large content
TS-50: Verify error handling if PDF generation fails

🔷 Module M9: Integration & End-to-End

TS-51: Verify complete workflow:
Input → API → AI → Output

TS-52: Verify data consistency across layers
TS-53: Verify system handles repeated usage without failure
TS-54: Verify no data loss during processing
TS-55: Verify integration under different inputs

🔷 Negative & Edge Case Testing

TS-56: Verify system behavior with empty input
TS-57: Verify extremely large input handling
TS-58: Verify invalid/special characters input
TS-59: Verify network interruption during request
TS-60: Verify system behavior when AI API is unavailable
TS-61: Verify system behavior under slow internet
TS-62: Verify duplicate submissions handling

🔷 Performance Testing

TS-63: Verify AI response time is within acceptable limit
TS-64: Verify system under multiple concurrent requests
TS-65: Verify no UI freezing during processing

🔷 Usability Testing

TS-66: Verify system is easy to use for non-technical users
TS-67: Verify labels and instructions are clear
TS-68: Verify user flow is intuitive

🔷 Regression Testing

TS-69: Verify existing features after bug fixes
TS-70: Verify no new issues introduced after changes